Queries for DBMS

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1. Create a database.

```
mysql> create database company
    -> ;
Query OK, 1 row affected (0.05 sec)
```

2. Use the same database.

```
mysql> use company;
Database changed
```

3. Create a table employee with fields/attributes “ emp_id, emp_name, designation, years_of_exp, age, city”.

```
mysql> create table employee(emp_id varchar(20),emp_name char(20),designation char(20),year_of_exp int,age int,city varchar(20))
    ->
    -> ;
Query OK, 0 rows affected (0.25 sec)
```

4. Insert 5 records into employee.

```
mysql> insert into employee values('BCADA31','rohan','Manager',2,20,'banglore');
Query OK, 1 row affected (0.32 sec)

mysql> insert into employee values('BCADA01','tony','Director',5,30,'Manglore');
Query OK, 1 row affected (0.17 sec)

mysql> insert into employee values('BCADA02','Cris','Analyst',6,33,'mysore');
Query OK, 1 row affected (0.05 sec)

mysql> insert into employee values('BCADA03','Angelina','associte',3,22,'udupi');
Query OK, 1 row affected (0.01 sec)

mysql> insert into employee values('BCADA04','Noah','Marketing-Head',5,29,'Chicago');
Query OK, 1 row affected (0.02 sec)
```

5. Display the data from employee table.

```
mysql> select * from employee;
+---------+----------+----------------+-------------+------+----------+
| emp_id  | emp_name | designation    | year_of_exp | age  | city     |
+---------+----------+----------------+-------------+------+----------+
| BCADA31 | rohan    | Manager        |           2 |   20 | banglore |
| BCADA01 | tony     | Director       |           5 |   30 | Manglore |
| BCADA02 | Cris     | Analyst        |           6 |   33 | mysore   |
| BCADA03 | Angelina | associte       |           3 |   22 | udupi    |
| BCADA04 | Noah     | Marketing-Head |           5 |   29 | Chicago  |
+---------+----------+----------------+-------------+------+----------+
5 rows in set (0.01 sec)
```

6. Display emp_id from employee table.

```
mysql> select emp_id from employee;
+---------+
| emp_id  |
+---------+
| BCADA31 |
| BCADA01 |
| BCADA02 |
| BCADA03 |
| BCADA04 |
+---------+
5 rows in set (0.00 sec)
```

7. Display emp_id and emp_name.

```
mysql> select emp_id,emp_name from employee;
+---------+----------+
| emp_id  | emp_name |
+---------+----------+
| BCADA31 | rohan    |
| BCADA01 | tony     |
| BCADA02 | Cris     |
| BCADA03 | Angelina |
| BCADA04 | Noah     |
+---------+----------+
5 rows in set (0.00 sec)
```

8. Display the details of employee where the designation is "Manager",

```
mysql> select * from employee where designation='manager';
+---------+----------+-------------+-------------+------+----------+
| emp_id  | emp_name | designation | year_of_exp | age  | city     |
+---------+----------+-------------+-------------+------+----------+
| BCADA31 | rohan    | Manager     |           2 |   20 | banglore |
+---------+----------+-------------+-------------+------+----------+
1 row in set (0.00 sec)
```

9. Display the emp_id and emp_name of "Bangalore" employees.

```
mysql> select emp_id,emp_name from employee where city='banglore';
+---------+----------+
| emp_id  | emp_name |
+---------+----------+
| BCADA31 | rohan    |
+---------+----------+
1 row in set (0.00 sec)
```

10. Display the name of employees with age < 35.

```
mysql> select emp_name from employee where age<35;
+----------+
| emp_name |
+----------+
| rohan    |
| tony     |
| Cris     |
| Angelina |
| Noah     |
+----------+
```

11. Display the employee data according to data name.

```
mysql> select * from employee order by emp_name;
+---------+----------+----------------+-------------+------+----------+
| emp_id  | emp_name | designation    | year_of_exp | age  | city     |
+---------+----------+----------------+-------------+------+----------+
| BCADA03 | Angelina | associte       |           3 |   22 | udupi    |
| BCADA02 | Cris     | Analyst        |           6 |   33 | mysore   |
| BCADA04 | Noah     | Marketing-Head |           5 |   29 | Chicago  |
| BCADA31 | rohan    | Manager        |           2 |   20 | banglore |
| BCADA01 | tony     | Director       |           5 |   30 | Manglore |
+---------+----------+----------------+-------------+------+----------+
5 rows in set (0.00 sec)
```

12. Sort the data in ascending order.

```
mysql> select * from employee order by emp_id;
+---------+----------+----------------+-------------+------+----------+
| emp_id  | emp_name | designation    | year_of_exp | age  | city     |
+---------+----------+----------------+-------------+------+----------+
| BCADA01 | tony     | Director       |           5 |   30 | Manglore |
| BCADA02 | Cris     | Analyst        |           6 |   33 | mysore   |
| BCADA03 | Angelina | associte       |           3 |   22 | udupi    |
| BCADA04 | Noah     | Marketing-Head |           5 |   29 | Chicago  |
| BCADA31 | rohan    | Manager        |           2 |   20 | banglore |
+---------+----------+----------------+-------------+------+----------+
5 rows in set (0.00 sec)
```

13. Create a table employee 3 with the same structure and data of the employee table.

```
mysql> create table employee_4 as select emp_name,designation from employee;
Query OK, 5 rows affected (0.04 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

14. Create a table employee 4 with the structure of employee and specific rows.

```
mysql> create table employee4 as select * from employee where year_of_exp<4;
Query OK, 2 rows affected (0.04 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> select * from employee4;
+---------+----------+-------------+-------------+------+----------+
| emp_id  | emp_name | designation | year_of_exp | age  | city     |
+---------+----------+-------------+-------------+------+----------+
| BCADA31 | rohan    | Manager     |           2 |   20 | banglore |
| BCADA03 | Angelina | associte    |           3 |   22 | udupi    |
+---------+----------+-------------+-------------+------+----------+
2 rows in set (0.00 sec)
```

15. Create a database student.

```
mysql> create database student;
Query OK, 1 row affected (0.01 sec)
```

16. Display the details of a student where the name is “Rohan” and the reg_no is 3.

```
mysql> select * from details where name = 'rohan' and reg_no=3;
+-------+--------+
| name  | reg_no |
+-------+--------+
| rohan |      3 |
| rohan |      3 |
+-------+--------+
2 rows in set (0.00 sec)
```

17. Find out if we can have multiple columns with distinct.

```
mysql> select distinct * from details;
+-------+--------+
| name  | reg_no |
+-------+--------+
| rohan |      3 |
| Tony  |      5 |
+-------+--------+
2 rows in set (0.00 sec)
```

19. Insert the entered values into capital letters(changing case).

```
mysql> select * from details where name = 'RoHan' and reg_no=3;
+-------+--------+
| name  | reg_no |
+-------+--------+
| rohan |      3 |
| rohan |      3 |
+-------+--------+
2 rows in set (0.00 sec)
```

20. Check whether the values entered into a table id case sensitive or not.

21. Find out whether 2 columns can be sorted together or not.

```
mysql> select * from details order by name,reg_no;
+-------+--------+
| name  | reg_no |
+-------+--------+
| rohan |      3 |
| rohan |      3 |
| Tony  |      5 |
+-------+--------+
3 rows in set (0.00 sec)
```

```
mysql> select * from details order by name,reg_no;
+-------+--------+
| name  | reg_no |
+-------+--------+
| noah  |      1 |
| rohan |      3 |
| rohan |      3 |
| Tony  |      5 |
+-------+--------+
4 rows in set (0.00 sec)
```

22. Find out whether we can sort the column values.

```
mysql> select * from details order by name;
+-------+--------+
| name  | reg_no |
+-------+--------+
| noah  |      1 |
| rohan |      3 |
| rohan |      3 |
| Tony  |      5 |
+-------+--------+
4 rows in set (0.00 sec)
```

23. Create a database SJU_student.

```
mysql> create database sju_student;
Query OK, 1 row affected (0.01 sec)
```

24. Use the same database.

```
mysql> use sju_student;
Database changed
```

25. Show all the databases created by the user.

```
mysql> show databases
    -> ;
+--------------------+
| Database           |
+--------------------+
| bank               |
| company            |
| information_schema |
| lab                |
| lab1               |
| lab2               |
| lab3               |
| lab_1              |
| lab_2              |
| laboratory_1       |
| mysql              |
| performance_schema |
| school             |
| sju_student        |
| stud               |
| student            |
| sys                |
+--------------------+
17 rows in set (0.01 sec)
```

26. Create a table student with fields/attributes " stud_id, stud_name, age, city, course, year(1,2,3), college_name, add_cred".

```
mysql> create table student(id varchar(20),name char(20),age int,city char(20),course char(20),year int,collage char(20),add_cred int);
Query OK, 0 rows affected (0.04 sec)
```

28. Display the structure of table.

```
mysql> desc student;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| id       | varchar(20) | YES  |     | NULL    |       |
| name     | char(20)    | YES  |     | NULL    |       |
| age      | int         | YES  |     | NULL    |       |
| city     | char(20)    | YES  |     | NULL    |       |
| course   | char(20)    | YES  |     | NULL    |       |
| year     | int         | YES  |     | NULL    |       |
| college  | char(20)    | YES  |     | NULL    |       |
| add_cred | int         | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
8 rows in set (0.01 sec)
```

29. Display the information of 2nd year students from the city “Mysore”.

```
mysql> select * from student where year=2 and city='mysore';
Empty set (0.00 sec)
```

30. Display the information of student where the age < 20 or year is 1st year.

```
mysql> select * from student where age<20 and year=1;
+------+-------+------+----------+--------+------+---------+----------+
| id   | name  | age  | city     | course | year | college | add_cred |
+------+-------+------+----------+--------+------+---------+----------+
| 2    | gagan |   19 | banglore | bcom   |    1 | sjcc    |        3 |
+------+-------+------+----------+--------+------+---------+----------+
1 row in set (0.00 sec)
```

31. Display the details of 1st year and 2nd year students.

```
mysql> select * from student where year in (1,2);
+------+-------+------+----------+--------+------+---------+----------+
| id   | name  | age  | city     | course | year | college | add_cred |
+------+-------+------+----------+--------+------+---------+----------+
| 1    | rohan |   20 | banglore | bcada  |    2 | sju     |        5 |
| 2    | gagan |   19 | banglore | bcom   |    1 | sjcc    |        3 |
| 4    | suraj |   21 | manglore | bba    |    2 | sjbit   |        1 |
+------+-------+------+----------+--------+------+---------+----------+
3 rows in set (0.00 sec)
```

32. Display the different types of city.

```
mysql> select distinct city from student ;
+----------+
| city     |
+----------+
| banglore |
| mysore   |
| manglore |
| Udupi    |
+----------+
4 rows in set (0.00 sec)
```

33. Display the stud_id, stud_name and age according to their name in the descending order.

```
mysql> select id,name,age from student order by name desc;
+------+--------+------+
| id   | name   | age  |
+------+--------+------+
| 3    | tarun  |   22 |
| 4    | suraj  |   21 |
| 1    | rohan  |   20 |
| 2    | gagan  |   19 |
| 5    | ashish |   18 |
+------+--------+------+
5 rows in set (0.00 sec)
```

34. Display the details of student except SJU.

```
mysql> select * from student where NOT COLLEGE='SJU';
+------+--------+------+----------+---------+------+---------+----------+
| id   | name   | age  | city     | course  | year | college | add_cred |
+------+--------+------+----------+---------+------+---------+----------+
| 2    | gagan  |   19 | banglore | bcom    |    1 | sjcc    |        3 |
| 3    | tarun  |   22 | mysore   | law     |    3 | sjcl    |        4 |
| 4    | suraj  |   21 | manglore | bba     |    2 | sjbit   |        1 |
| 5    | ashish |   18 | Udupi    | bcomifa |    3 | ramahia |        3 |
+------+--------+------+----------+---------+------+---------+----------+
4 rows in set (0.00 sec)
```

35. Find orderby for all the columns

```
mysql> select * from student order by name;
+------+--------+------+----------+---------+------+---------+----------+
| id   | name   | age  | city     | course  | year | college | add_cred |
+------+--------+------+----------+---------+------+---------+----------+
| 5    | ashish |   18 | Udupi    | bcomifa |    3 | ramahia |        3 |
| 2    | gagan  |   19 | banglore | bcom    |    1 | sjcc    |        3 |
| 1    | rohan  |   20 | banglore | bcada   |    2 | sju     |        5 |
| 4    | suraj  |   21 | manglore | bba     |    2 | sjbit   |        1 |
| 3    | tarun  |   22 | mysore   | law     |    3 | sjcl    |        4 |
+------+--------+------+----------+---------+------+---------+----------+
5 rows in set (0.00 sec)

mysql> select * from student order by age;
+------+--------+------+----------+---------+------+---------+----------+
| id   | name   | age  | city     | course  | year | college | add_cred |
+------+--------+------+----------+---------+------+---------+----------+
| 5    | ashish |   18 | Udupi    | bcomifa |    3 | ramahia |        3 |
| 2    | gagan  |   19 | banglore | bcom    |    1 | sjcc    |        3 |
| 1    | rohan  |   20 | banglore | bcada   |    2 | sju     |        5 |
| 4    | suraj  |   21 | manglore | bba     |    2 | sjbit   |        1 |
| 3    | tarun  |   22 | mysore   | law     |    3 | sjcl    |        4 |
+------+--------+------+----------+---------+------+---------+----------+
5 rows in set (0.00 sec)

mysql> select * from student order by city;
+------+--------+------+----------+---------+------+---------+----------+
| id   | name   | age  | city     | course  | year | college | add_cred |
+------+--------+------+----------+---------+------+---------+----------+
| 1    | rohan  |   20 | banglore | bcada   |    2 | sju     |        5 |
| 2    | gagan  |   19 | banglore | bcom    |    1 | sjcc    |        3 |
| 4    | suraj  |   21 | manglore | bba     |    2 | sjbit   |        1 |
| 3    | tarun  |   22 | mysore   | law     |    3 | sjcl    |        4 |
| 5    | ashish |   18 | Udupi    | bcomifa |    3 | ramahia |        3 |
+------+--------+------+----------+---------+------+---------+----------+
5 rows in set (0.00 sec)

mysql> select * from student order by course;
+------+--------+------+----------+---------+------+---------+----------+
| id   | name   | age  | city     | course  | year | college | add_cred |
+------+--------+------+----------+---------+------+---------+----------+
| 4    | suraj  |   21 | manglore | bba     |    2 | sjbit   |        1 |
| 1    | rohan  |   20 | banglore | bcada   |    2 | sju     |        5 |
| 2    | gagan  |   19 | banglore | bcom    |    1 | sjcc    |        3 |
| 5    | ashish |   18 | Udupi    | bcomifa |    3 | ramahia |        3 |
| 3    | tarun  |   22 | mysore   | law     |    3 | sjcl    |        4 |
+------+--------+------+----------+---------+------+---------+----------+
5 rows in set (0.00 sec)

mysql> select * from student order by year;
+------+--------+------+----------+---------+------+---------+----------+
| id   | name   | age  | city     | course  | year | college | add_cred |
+------+--------+------+----------+---------+------+---------+----------+
| 2    | gagan  |   19 | banglore | bcom    |    1 | sjcc    |        3 |
| 1    | rohan  |   20 | banglore | bcada   |    2 | sju     |        5 |
| 4    | suraj  |   21 | manglore | bba     |    2 | sjbit   |        1 |
| 3    | tarun  |   22 | mysore   | law     |    3 | sjcl    |        4 |
| 5    | ashish |   18 | Udupi    | bcomifa |    3 | ramahia |        3 |
+------+--------+------+----------+---------+------+---------+----------+
5 rows in set (0.00 sec)
```

19-12-2024

1. Show all existing

```
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| lab_1              |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.34 sec)
```

2. Create a database student_04 and use the same database

```
mysql> create database student_231bcada31;
Query OK, 1 row affected (0.14 sec)

mysql> use student_231bcada31;
Database changed
mysql>
```

3. Create a table student with id , name, year, department, college and city ,insert 5 values

```
mysql> create table student_231bcada31(id int,name varchar(20),year int, department varchar(20),collage varchar(20), city varchar(20));
Query OK, 0 rows affected (0.57 sec)
```

```
mysql> insert into student_231bcada31 values(1,'rohan',2,'BigData','sju','banglore'),
    -> (2,'rekha',3,'CS','sjcc','mumbai'),
    -> (3,'gagan',1,"Commerce",'jain','chennai'),
    -> (4,'swathi',1,'humanity','Jyothi','delhi'),
    -> (5,'anitha',2,'computers','mcc','banglore');
Query OK, 5 rows affected (0.14 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

4. Display all the details

```
mysql> select * from student_231bcada31;
+------+--------+------+------------+---------+----------+
| id   | name   | year | department | collage | city     |
+------+--------+------+------------+---------+----------+
|    1 | rohan  |    2 | BigData    | sju     | banglore |
|    2 | rekha  |    3 | CS         | sjcc    | mumbai   |
|    3 | gagan  |    1 | Commerce   | jain    | chennai  |
|    4 | swathi |    1 | humanity   | Jyothi  | delhi    |
|    5 | anitha |    2 | computers  | mcc     | banglore |
+------+--------+------+------------+---------+----------+
5 rows in set (0.00 sec)
```

5. update the college names to sju and display all details

```
mysql> update student_231bcada31 set collage= 'sju';
Query OK, 4 rows affected (0.09 sec)
Rows matched: 5  Changed: 4  Warnings: 0
```

6. Update city into Bengaluru where city is Bangalore and then display

```
mysql> update student_231bcada31 set city='bengalure' where city='banglore';
Query OK, 2 rows affected (0.13 sec)
Rows matched: 2  Changed: 2  Warnings: 0

mysql> select * from student_231bcada31;
+------+--------+------+------------+---------+-----------+
| id   | name   | year | department | collage | city      |
+------+--------+------+------------+---------+-----------+
|    1 | rohan  |    2 | BigData    | sju     | bengalure |
|    2 | rekha  |    3 | CS         | sju     | mumbai    |
|    3 | gagan  |    1 | Commerce   | sju     | chennai   |
|    4 | swathi |    1 | humanity   | sju     | delhi     |
|    5 | anitha |    2 | computers  | sju     | bengalure |
+------+--------+------+------------+---------+-----------+
5 rows in set (0.00 sec)
```

7.

a. Display all the details of student whose name starts with a
b. Name ends with a
c. 1st and last letter is a
d. 2nd character is a
e.
f. 3rd character is e

g. Name with 6 characters

```
mysql> select * from student_231bcada31;
+------+--------+------+------------+---------+-----------+
| id   | name   | year | department | collage | city      |
+------+--------+------+------------+---------+-----------+
|    1 | rohan  |    2 | BigData    | sju     | bengalure |
|    2 | rekha  |    3 | CS         | sju     | mumbai    |
|    3 | gagan  |    1 | Commerce   | sju     | chennai   |
|    4 | swathi |    1 | humanity   | sju     | delhi     |
|    5 | anitha |    2 | computers  | sju     | bengalure |
+------+--------+------+------------+---------+-----------+
5 rows in set (0.00 sec)

mysql> select * from student_231bcada31 where name like 'a%';
+------+--------+------+------------+---------+-----------+
| id   | name   | year | department | collage | city      |
+------+--------+------+------------+---------+-----------+
|    5 | anitha |    2 | computers  | sju     | bengalure |
+------+--------+------+------------+---------+-----------+
1 row in set (0.04 sec)

mysql> select * from student_231bcada31 where name like 'a%a';
+------+--------+------+------------+---------+-----------+
| id   | name   | year | department | collage | city      |
+------+--------+------+------------+---------+-----------+
|    5 | anitha |    2 | computers  | sju     | bengalure |
+------+--------+------+------------+---------+-----------+
1 row in set (0.00 sec)
```

```
mysql> select * from student_231bcada31 where name like '_a%';
+------+-------+------+------------+---------+---------+
| id   | name  | year | department | collage | city    |
+------+-------+------+------------+---------+---------+
|    3 | gagan |    1 | Commerce   | sju     | chennai |
+------+-------+------+------------+---------+---------+
1 row in set (0.00 sec)

mysql> select * from student_231bcada31 where name like '_e%';
+------+-------+------+------------+---------+--------+
| id   | name  | year | department | collage | city   |
+------+-------+------+------------+---------+--------+
|    2 | rekha |    3 | CS         | sju     | mumbai |
+------+-------+------+------------+---------+--------+
1 row in set (0.00 sec)

mysql> select * from student_231bcada31 where name like '______';
+------+--------+------+------------+---------+-----------+
| id   | name   | year | department | collage | city      |
+------+--------+------+------------+---------+-----------+
|    4 | swathi |    1 | humanity   | sju     | delhi     |
|    5 | anitha |    2 | computers  | sju     | bengalure |
+------+--------+------+------------+---------+-----------+
2 rows in set (0.00 sec)
```

8. Adding a new column country and then dis

```
mysql> alter table student_231bcada31 add column country varchar(20);
Query OK, 0 rows affected (0.29 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select * from student_231bcada31 ;
+------+--------+------+------------+---------+-----------+---------+
| id   | name   | year | department | collage | city      | country |
+------+--------+------+------------+---------+-----------+---------+
|    1 | rohan  |    2 | BigData    | sju     | bengalure | NULL    |
|    2 | rekha  |    3 | CS         | sju     | mumbai    | NULL    |
|    3 | gagan  |    1 | Commerce   | sju     | chennai   | NULL    |
|    4 | swathi |    1 | humanity   | sju     | delhi     | NULL    |
|    5 | anitha |    2 | computers  | sju     | bengalure | NULL    |
+------+--------+------+------------+---------+-----------+---------+
5 rows in set (0.00 sec)
```

play

9. Display the structure of the table

```
mysql> desc student_231bcada31;
+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| id           | int         | YES  |     | NULL    |       |
| student_name | varchar(20) | YES  |     | NULL    |       |
| year         | int         | YES  |     | NULL    |       |
| department   | varchar(20) | YES  |     | NULL    |       |
| collage      | varchar(20) | YES  |     | NULL    |       |
| city         | varchar(20) | YES  |     | NULL    |       |
| country      | varchar(20) | YES  |     | NULL    |       |
+--------------+-------------+------+-----+---------+-------+
7 rows in set (0.00 sec)
```

10. Update the values for country column to india and display

```
mysql> select * from student_231bcada31;
+------+--------+------+------------+---------+-----------+---------+
| id   | name   | year | department | collage | city      | country |
+------+--------+------+------------+---------+-----------+---------+
|    1 | rohan  |    2 | BigData    | sju     | bengalure | india   |
|    2 | rekha  |    3 | CS         | sju     | mumbai    | india   |
|    3 | gagan  |    1 | Commerce   | sju     | chennai   | india   |
|    4 | swathi |    1 | humanity   | sju     | delhi     | india   |
|    5 | anitha |    2 | computers  | sju     | bengalure | india   |
+------+--------+------+------------+---------+-----------+---------+
5 rows in set (0.00 sec)
```

11. Rename a column name

```
mysql> alter table student_231bcada31 rename column name to student_name;
Query OK, 0 rows affected (0.18 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

9-01-2025

Q1) Show all the databases created by the user.

```
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| lab_1              |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.01 sec)
```

Q2) Create new database lab1.

```
mysql> create database lab_1;
Query OK, 1 row affected (0.01 sec)
```

```
mysql> use lab_1;
Database changed
```

Q3) Create table employee_231BCADA31(your reg. no.) with the following attributes.

```
mysql>  create table employee_231bcada31(id varchar(10),name varchar(25),DOB Date,
ty varchar(20),gender char(8));
Query OK, 0 rows affected (0.01 sec)
```

Q4) Insert 10 records.

```
mysql> insert into employee_231bcada31 values  ("E126","Ananya","1993-01-30","CEO","C-Suit",57000,"Chennai","Female")
    -> ("E127","Bhavya","1994-01-30","CMO","Marketing",28000,"Chitradurga","Female"),
    ->  ('E128',"Akhil","1995-01-30","Ass. Vice President","Corporation",29000,"Manglore","Male"),
    -> ("E129","Nishitha","1996-01-30","MD","Management",60000,"Udupi","Female"),
    ->  ("E130","Jack","1997-01-30","CFO","Accounts",75000,"Davagere","Male"),
    -> ("E131","Peter","1998-01-30","CTO","IT",34000,"Chittur","Male"),
    ->  ("E132","David","1999-01-30","HOD","PR",45000,"Gujarat","Male");
Query OK, 7 rows affected (0.00 sec)
Records: 7  Duplicates: 0  Warnings: 0
```

```
mysql> insert into employee_231bcada31(id,name,DOB,designation,department,salary,city,gender) values ('E124','Rohan','1991-01-30',"Dept. Manager","H
R",70000,"Mysore","Male"),
    ->  ("E125","Ramya","1992-01-30","Vice President","Corporation",72000,"Hassan","Female"),
    ->  ("E125","Ramya","1992-01-30","Vice President","Corporation",72000,"Hassan","Female");
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

Q5) Display the data.

```
mysql> select * from employee_231bcada31 order by id desc;
+------+----------+------------+---------------------+-------------+----------+-------------+--------+---------+
| id   | name     | DOB        | designation         | department  | salary   | city        | gender | pincode |
+------+----------+------------+---------------------+-------------+----------+-------------+--------+---------+
| E132 | David    | 1999-01-30 | HOD                 | PR          | 45000.00 | Gujarat     | Male   | 12      |
| E131 | Peter    | 1998-01-30 | CTO                 | IT          | 34000.00 | Chittur     | Male   | 76      |
| E130 | Jack     | 1997-01-30 | CFO                 | Accounts    | 75000.00 | Davagere    | Male   | 11      |
| E129 | Nishitha | 1996-01-30 | MD                  | Management  | 60000.00 | Udupi       | Female | 56      |
| E128 | Akhil    | 1995-01-30 | Ass. Vice President | Corporation | 29000.00 | Manglore    | Male   | 46      |
| E127 | Bhavya   | 1994-01-30 | CMO                 | Marketing   | 28000.00 | Chitradurga | Female | 36      |
| E126 | Ananya   | 1993-01-30 | CEO                 | C-Suit      | 57000.00 | Chennai     | Female | 26      |
| E125 | Ramya    | 1992-01-30 | Vice President      | Corporation | 72000.00 | Hassan      | Female | 16      |
| E125 | Ramya    | 1992-01-30 | Vice President      | Corporation | 72000.00 | Hassan      | Female | 16      |
| E124 | Rohan    | 1991-01-30 | Dept. Manager       | HR          | 70000.00 | Mysore      | Male   | 6       |
| E123 | Rahul    | 1990-01-30 | Manager             | HR          | 80000.00 | Bengaluru   | Male   | 56      |
+------+----------+------------+---------------------+-------------+----------+-------------+--------+---------+
11 rows in set (0.00 sec)
```

Q6) Display the structure of the table.

```
mysql> desc employee_231bcada31;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| id          | varchar(10)  | YES  |     | NULL    |       |
| name        | varchar(25)  | YES  |     | NULL    |       |
| DOB         | date         | YES  |     | NULL    |       |
| designation | varchar(30)  | YES  |     | NULL    |       |
| department  | char(30)     | YES  |     | NULL    |       |
| salary      | decimal(7,2) | YES  |     | NULL    |       |
| city        | varchar(20)  | YES  |     | NULL    |       |
| gender      | char(8)      | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
8 rows in set (0.01 sec)
```

Q7) Add a column pincode and insert values in it.

```
mysql> alter table employee_231bcada31 add pincode varchar(20);
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> update employee_231bcada31 set pincode= case
    -> when city = "bengaluru" then 56
    -> when city = "Mysore" then 06
    -> when city = "chittur" then 76
    -> when city = "davagiri" then 66
    -> when city = "udupi" then 56
    -> when city = "manglore" then 46
    -> when city = "chitradurga" then 36
    -> when city = "chennai" then 26
    -> when city = "hassan" then 16
    -> end;
Query OK, 9 rows affected (0.01 sec)
Rows matched: 11  Changed: 9  Warnings: 0
```

Q8) Display the different designation.

```
mysql>  select distinct designation from employee_231bcada31;
+---------------------+
| designation         |
+---------------------+
| Manager             |
| Dept. Manager       |
| Vice President      |
| CEO                 |
| CMO                 |
| Ass. Vice President |
| MD                  |
| CFO                 |
| CTO                 |
| HOD                 |
+---------------------+
10 rows in set (0.00 sec)
```

Q9) Arrange data according to empid , descending order.

mysql> select * from employee_231bcada31 order by id desc;

id	name	DOB	designation	department	salary	city	gender	pincode
E132	David	1999-01-30	HOD	PR	45000.00	Gujarat	Male	12
E131	Peter	1998-01-30	CTO	IT	34000.00	Chittur	Male	76
E130	Jack	1997-01-30	CFO	Accounts	75000.00	Davagere	Male	11
E129	Nishitha	1996-01-30	MD	Management	60000.00	Udupi	Female	56
E128	Akhil	1995-01-30	Ass. Vice President	Corporation	29000.00	Manglore	Male	46
E127	Bhavya	1994-01-30	CMO	Marketing	28000.00	Chitradurga	Female	36
E126	Ananya	1993-01-30	CEO	C-Suit	57000.00	Chennai	Female	26
E125	Ramya	1992-01-30	Vice President	Corporation	72000.00	Hassan	Female	16
E125	Ramya	1992-01-30	Vice President	Corporation	72000.00	Hassan	Female	16
E124	Rohan	1991-01-30	Dept. Manager	HR	70000.00	Mysore	Male	6
E123	Rahul	1990-01-30	Manager	HR	80000.00	Bengaluru	Male	56

11 rows in set (0.00 sec)

Q10) Arrange the data according to names of the employee ascending order.

mysql> select * from employee_231bcada31 order by name;

id	name	DOB	designation	department	salary	city	gender
E128	Akhil	1995-01-30	Ass. Vice President	Corporation	29000.00	Manglore	Male
E126	Ananya	1993-01-30	CEO	C-Suit	57000.00	Chennai	Female
E127	Bhavya	1994-01-30	CMO	Marketing	28000.00	Chitradurga	Female
E132	David	1999-01-30	HOD	PR	45000.00	Gujarat	Male
E130	Jack	1997-01-30	CFO	Accounts	75000.00	Davagere	Male
E129	Nishitha	1996-01-30	MD	Management	60000.00	Udupi	Female
E131	Peter	1998-01-30	CTO	IT	34000.00	Chittur	Male
E123	Rahul	1990-01-30	Manager	HR	80000.00	Bengaluru	Male
E125	Ramya	1992-01-30	Vice President	Corporation	72000.00	Hassan	Female
E124	Rohan	1991-01-30	Dept. Manager	HR	70000.00	Mysore	Male

10 rows in set (0.05 sec)

Q11) Display the minimum salary and maximum salary.

```
mysql>  select min(salary) from employee_231BCADA31;
+-------------+
| min(salary) |
+-------------+
|    28000.00 |
+-------------+
1 row in set (0.08 sec)

mysql>  select max(salary) from employee_231BCADA31;
+-------------+
| max(salary) |
+-------------+
|    80000.00 |
+-------------+
1 row in set (0.00 sec)
```

Q12) Update salary to 75,000 if the department is accounts and the post is deputy manager.

```
mysql> update employee_231bcada31 set salary=75000 where department="accounts" and designation="dept. manager";
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0
```

Q13) Display the information of employee where the salary is less than 30,000.

```
mysql> select * from employee_231bcada31 where salary<30000;
+------+--------+------------+---------------------+-------------+----------+-------------+--------+
| id   | name   | DOB        | designation         | department  | salary   | city        | gender |
+------+--------+------------+---------------------+-------------+----------+-------------+--------+
| E127 | Bhavya | 1994-01-30 | CMO                 | Marketing   | 28000.00 | Chitradurga | Female |
| E128 | Akhil  | 1995-01-30 | Ass. Vice President | Corporation | 29000.00 | Manglore    | Male   |
+------+--------+------------+---------------------+-------------+----------+-------------+--------+
2 rows in set (0.00 sec)
```

Q14) Update the city to Bengaluru and department to marketing if the empid is either 124 or 125.

```
mysql> UPDATE employee_231bcada31 set city="bengaluru",department="marketing" where id= "E124" or id="E125"
    -> ;
Query OK, 2 rows affected (0.16 sec)
Rows matched: 2  Changed: 2  Warnings: 0

mysql> select * from employee_231bcada31;
+------+----------+------------+---------------------+-------------+----------+-------------+--------+
| id   | name     | DOB        | designation         | department  | salary   | city        | gender |
+------+----------+------------+---------------------+-------------+----------+-------------+--------+
| E123 | Rahul    | 1990-01-30 | Manager             | HR          | 80000.00 | Bengaluru   | Male   |
| E124 | Rohan    | 1991-01-30 | Dept. Manager       | marketing   | 70000.00 | bengaluru   | Male   |
| E125 | Ramya    | 1992-01-30 | Vice President      | marketing   | 72000.00 | bengaluru   | Female |
| E126 | Ananya   | 1993-01-30 | CEO                 | C-Suit      | 57000.00 | Chennai     | Female |
| E127 | Bhavya   | 1994-01-30 | CMO                 | Marketing   | 28000.00 | Chitradurga | Female |
| E128 | Akhil    | 1995-01-30 | Ass. Vice President | Corporation | 29000.00 | Manglore    | Male   |
| E129 | Nishitha | 1996-01-30 | MD                  | Management  | 60000.00 | Udupi       | Female |
| E130 | Jack     | 1997-01-30 | CFO                 | Accounts    | 75000.00 | Davagere    | Male   |
| E131 | Peter    | 1998-01-30 | CTO                 | IT          | 34000.00 | Chittur     | Male   |
| E132 | David    | 1999-01-30 | HOD                 | PR          | 45000.00 | Gujarat     | Male   |
+------+----------+------------+---------------------+-------------+----------+-------------+--------+
10 rows in set (0.00 sec)
```

Q15) Create table employee1 with the same structure and data of employee.

```
mysql> CREATE table employee1 as select * from employee_231bcada31;
Query OK, 10 rows affected (0.30 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

Q16) Create table employee2 with only structure as employee.

```
mysql> CREATE table employee2 as select * from employee_231bcada31 where 1=0;
Query OK, 0 rows affected (0.74 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Q17) Display names of employee where the name starts from a.

```
mysql> select name from employee_231bcada31 where name like "a%"
    -> ;
+--------+
| name   |
+--------+
| Ananya |
| Akhil  |
+--------+
2 rows in set (0.03 sec)
```

Q18) Display empid and ename who belongs to a city mysore, Bengaluru and hassan.

```
mysql> select id,name from employee_231bcada31 where city in ("bengaluru","mysore","hassan");
+------+-------+
| id   | name  |
+------+-------+
| E123 | Rahul |
| E124 | Rohan |
| E125 | Ramya |
+------+-------+
3 rows in set (0.05 sec)
```

Q19) Demonstrate not in clause.

```
mysql> select id,name from employee_231bcada31 where designation not in ("Manager","ass. vice president","dept. manager");
+------+----------+
| id   | name     |
+------+----------+
| E125 | Ramya    |
| E126 | Ananya   |
| E127 | Bhavya   |
| E129 | Nishitha |
| E130 | Jack     |
| E131 | Peter    |
| E132 | David    |
+------+----------+
7 rows in set (0.00 sec)
```

Q20) Demonstrate and,or,not .

```
mysql> select * from employee_231bcada31 where not salary>=50000 ;
+------+--------+------------+---------------------+-------------+----------+-------------+--------+
| id   | name   | DOB        | designation         | department  | salary   | city        | gender |
+------+--------+------------+---------------------+-------------+----------+-------------+--------+
| E127 | Bhavya | 1994-01-30 | CMO                 | Marketing   | 28000.00 | Chitradurga | Female |
| E128 | Akhil  | 1995-01-30 | Ass. Vice President | Corporation | 29000.00 | Manglore    | Male   |
| E131 | Peter  | 1998-01-30 | CTO                 | IT          | 34000.00 | Chittur     | Male   |
| E132 | David  | 1999-01-30 | HOD                 | PR          | 45000.00 | Gujarat     | Male   |
+------+--------+------------+---------------------+-------------+----------+-------------+--------+
4 rows in set (0.03 sec)
```

```
ysql> select name,id,salary from employee_231bcada31 where department="hr" and city="bengaluru";
-------+------+----------+
 name  | id   | salary   |
-------+------+----------+
 Rahul | E123 | 80000.00 |
-------+------+----------+
 row in set (0.00 sec)
```

```
mysql> select * from employee_231bcada31 where designation ="manager" or designation = "dept. manager" ;
+------+-------+------------+---------------+------------+----------+-----------+--------+
| id   | name  | DOB        | designation   | department | salary   | city      | gender |
+------+-------+------------+---------------+------------+----------+-----------+--------+
| E123 | Rahul | 1990-01-30 | Manager       | HR         | 80000.00 | Bengaluru | Male   |
| E124 | Rohan | 1991-01-30 | Dept. Manager | marketing  | 70000.00 | bengaluru | Male   |
+------+-------+------------+---------------+------------+----------+-----------+--------+
2 rows in set (0.00 sec)
```

Q21) Display the ename where the salary ranges between 50,000 to 75,000.

```
mysql> select name from employee_231bcada31 where salary between 50000 and  70000;
+----------+
| name     |
+----------+
| Rohan    |
| Ananya   |
| Nishitha |
+----------+
3 rows in set (0.05 sec)
```

Q22) Delete the data from employee1 where designation is manager.

```
mysql> delete from employee1 where designation='manager';
Query OK, 1 row affected (0.13 sec)

mysql> select * from employee1
    -> ;
+------+----------+------------+--------------------+-------------+----------+-------------+--------+
| id   | name     | DOB        | designation        | department  | salary   | city        | gender |
+------+----------+------------+--------------------+-------------+----------+-------------+--------+
| E124 | Rohan    | 1991-01-30 | Dept. Manager      | marketing   | 70000.00 | bengaluru   | Male   |
| E125 | Ramya    | 1992-01-30 | Vice President     | marketing   | 72000.00 | bengaluru   | Female |
| E126 | Ananya   | 1993-01-30 | CEO                | C-Suit      | 57000.00 | Chennai     | Female |
| E127 | Bhavya   | 1994-01-30 | CMO                | Marketing   | 28000.00 | Chitradurga | Female |
| E128 | Akhil    | 1995-01-30 | Ass. Vice President| Corporation | 29000.00 | Manglore    | Male   |
| E129 | Nishitha | 1996-01-30 | MD                 | Management  | 60000.00 | Udupi       | Female |
| E130 | Jack     | 1997-01-30 | CFO                | Accounts    | 75000.00 | Davagere    | Male   |
| E131 | Peter    | 1998-01-30 | CTO                | IT          | 34000.00 | Chittur     | Male   |
| E132 | David    | 1999-01-30 | HOD                | PR          | 45000.00 | Gujarat     | Male   |
+------+----------+------------+--------------------+-------------+----------+-------------+--------+
9 rows in set (0.00 sec)
```

Q23) Drop the table employee2.

```
mysql> drop table  employee2;
Query OK, 0 rows affected (0.49 sec)
```

Q24) Display top 3 salaries along with ename(limit).

```
mysql> select name,salary from employee_231bcada31 order by salary limit 3;
+--------+----------+
| name   | salary   |
+--------+----------+
| Bhavya | 28000.00 |
| Akhil  | 29000.00 |
| Peter  | 34000.00 |
+--------+----------+
3 rows in set (0.00 sec)

mysql> select name,salary from employee_231bcada31 order by salary desc  limit 3;
+-------+----------+
| name  | salary   |
+-------+----------+
| Rahul | 80000.00 |
| Jack  | 75000.00 |
| Ramya | 72000.00 |
+-------+----------+
3 rows in set (0.00 sec)
```

Q25) Give the bonus to employee(10% of their salary) where the department is maintenance.

```
mysql> UPDATE employee_231bcada31 set salary=salary+(salary*0.10) where department="management";
Query OK, 1 row affected (0.10 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

Q26) Calculate the age of employee.

```
mysql> select *,timestampdiff(year,dob,curdate()) as age from employee_231bcada31;
+------+----------+------------+----------------------+-------------+----------+-------------+--------+------+
| id   | name     | DOB        | designation          | department  | salary   | city        | gender | age  |
+------+----------+------------+----------------------+-------------+----------+-------------+--------+------+
| E123 | Rahul    | 1990-01-30 | Manager              | HR          | 80000.00 | Bengaluru   | Male   |   34 |
| E124 | Rohan    | 1991-01-30 | Dept. Manager        | marketing   | 70000.00 | bengaluru   | Male   |   33 |
| E125 | Ramya    | 1992-01-30 | Vice President       | marketing   | 72000.00 | bengaluru   | Female |   32 |
| E126 | Ananya   | 1993-01-30 | CEO                  | C-Suit      | 57000.00 | Chennai     | Female |   31 |
| E127 | Bhavya   | 1994-01-30 | CMO                  | Marketing   | 28000.00 | Chitradurga | Female |   30 |
| E128 | Akhil    | 1995-01-30 | Ass. Vice President  | Corporation | 29000.00 | Manglore    | Male   |   29 |
| E129 | Nishitha | 1996-01-30 | MD                   | Management  | 66000.00 | Udupi       | Female |   28 |
| E130 | Jack     | 1997-01-30 | CFO                  | Accounts    | 75000.00 | Davagere    | Male   |   27 |
| E131 | Peter    | 1998-01-30 | CTO                  | IT          | 34000.00 | Chittur     | Male   |   26 |
| E132 | David    | 1999-01-30 | HOD                  | PR          | 45000.00 | Gujarat     | Male   |   25 |
+------+----------+------------+----------------------+-------------+----------+-------------+--------+------+
10 rows in set (0.00 sec)
```

24-1-25

1. Create database

```
mysql> create database lab1_31;
Query OK, 1 row affected (0.09 sec)

mysql> use lab1_31;
Database changed
```

2. Create table

```
mysql> create table 231bcada31(id varchar(20),name varchar(20),design varchar(20),city varchar(20));
Query OK, 0 rows affected (0.42 sec)
```

3. Insert values

```
mysql> insert into 231bcada31 values(31,'rohan','manager','banglore');
Query OK, 1 row affected (0.09 sec)

mysql> insert into 231bcada31 values(32,'rekhashree','assistant manager','mumbai');
Query OK, 1 row affected (0.10 sec)
```

4. Display table

```
mysql> select * from 231bcada31;
+------+------------+-------------------+----------+
| id   | name       | design            | city     |
+------+------------+-------------------+----------+
| 31   | rohan      | manager           | banglore |
| 32   | rekhashree | assistant manager | mumbai   |
| 33   | gagan      | director          | delhi    |
+------+------------+-------------------+----------+
3 rows in set (0.00 sec)
```

5. Update city from Bangalore to Bengaluru

```
mysql> update 231bcada31 set city = 'bengalure' where city='banglore';
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql> select * from 231bcada31;
+------+------------+-------------------+-----------+
| id   | name       | design            | city      |
+------+------------+-------------------+-----------+
| 31   | rohan      | manager           | bengalure |
| 32   | rekhashree | assistant manager | mumbai    |
| 33   | gagan      | director          | delhi     |
+------+------------+-------------------+-----------+
3 rows in set (0.00 sec)
```

6. Add a column salary

```
mysql> alter table 231bcada31 add column salary decimal;
Query OK, 0 rows affected (0.14 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select * from 231bcada31;
+------+------------+-------------------+-----------+--------+
| id   | name       | design            | city      | salary |
+------+------------+-------------------+-----------+--------+
| 31   | rohan      | manager           | bengalure |   NULL |
| 32   | rekhashree | assistant manager | mumbai    |   NULL |
| 33   | gagan      | director          | delhi     |   NULL |
+------+------------+-------------------+-----------+--------+
3 rows in set (0.00 sec)
```

7. Insert values into salary column

```
mysql> update 231bcada31 set salary=case
    -> when design= 'manager' then 20000
    -> when design= 'assistant manager' then 30000
    -> when design= 'director' then 15000
    -> end;
Query OK, 3 rows affected (0.08 sec)
Rows matched: 3  Changed: 3  Warnings: 0

mysql> select * from 231bcada31;
+------+------------+-------------------+-----------+--------+
| id   | name       | design            | city      | salary |
+------+------------+-------------------+-----------+--------+
| 31   | rohan      | manager           | bengalure |  20000 |
| 32   | rekhashree | assistant manager | mumbai    |  30000 |
| 33   | gagan      | director          | delhi     |  15000 |
+------+------------+-------------------+-----------+--------+
3 rows in set (0.00 sec)
```

8. Rename a column from design to designation

```
mysql> alter table 231bcada31 rename column design to designation;
Query OK, 0 rows affected (0.14 sec)
Records: 0  Duplicates: 0  Warnings: 0
mysql> alter table 231bcada31 modify designation char(30);
Query OK, 3 rows affected (0.88 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

9. Modify datatype of designation to char

```
mysql> alter table 231bcada31 modify designation char(30);
Query OK, 3 rows affected (0.88 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

10. Drop country column

```
mysql> alter table 231bcada31 drop city;
Query OK, 0 rows affected (0.19 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> select * from 231bcada31;
+------+------------+-------------------+--------+
| id   | name       | designation       | salary |
+------+------------+-------------------+--------+
| 31   | rohan      | manager           |  20000 |
| 32   | rekhashree | assistant manager |  30000 |
| 33   | gagan      | director          |  15000 |
+------+------------+-------------------+--------+
3 rows in set (0.00 sec)
```

Aggregation function

1. Create table

```
mysql> create table student_231bcada31 (name varchar(20),regno varchar(20),marks int);
Query OK, 0 rows affected (0.21 sec)
```

2. Insert values

```
mysql> INSERT INTO student_231bcada31 (name, regno, marks) VALUES
    -> ('rahul', 31, 90),
    -> ('rohan', 32, 85),
    -> ('rajesh', 33, 95),
    -> ('rohith', 34, 88);
Query OK, 4 rows affected (0.09 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

3. Display

```
mysql> select * from student_231bcada31;
+--------+-------+-------+
| name   | regno | marks |
+--------+-------+-------+
| rahul  | 31    |    90 |
| rohan  | 32    |    85 |
| rajesh | 33    |    95 |
| rohith | 34    |    88 |
+--------+-------+-------+
4 rows in set (0.00 sec)
```

4. Insert value only in reg no and marks

```
mysql> INSERT INTO student_231bcada31 (regno, marks) VALUES(35,97);
Query OK, 1 row affected (0.09 sec)

mysql> select * from student_231bcada31;
+--------+-------+-------+
| name   | regno | marks |
+--------+-------+-------+
| rahul  | 31    |    90 |
| rohan  | 32    |    85 |
| rajesh | 33    |    95 |
| rohith | 34    |    88 |
| NULL   | 35    |    97 |
+--------+-------+-------+
5 rows in set (0.00 sec)
```

5. Display max marks

```
mysql> select max(marks) from student_231bcada31;
+------------+
| max(marks) |
+------------+
|         97 |
+------------+
1 row in set (0.12 sec)
```

6. Max marks with column name as highest marks

```
mysql> select max(marks) 'Highest Marks'  from student_231bcada31;
+---------------+
| Highest Marks |
+---------------+
|            97 |
+---------------+
1 row in set (0.00 sec)
```

7. Min marks with column name as lowest marks

```
mysql> select min(marks) 'lowest Marks'  from student_231bcada31;
+--------------+
| lowest Marks |
+--------------+
|           85 |
+--------------+
1 row in set (0.00 sec)
```

8. Sum of marks with column name as total

```
mysql> select sum(marks) 'Total'  from student_231bcada31;
+-------+
| Total |
+-------+
|   455 |
+-------+
1 row in set (0.02 sec)
```

9. Avg of marks with column name as average

```
mysql> select avg(marks)   from student_231bcada31;
+------------+
| avg(marks) |
+------------+
|    91.0000 |
+------------+
1 row in set (0.00 sec)
```

10. Count of students table [no.of rows]

```
mysql> select count(*)   from student_231bcada31;
+----------+
| count(*) |
+----------+
|        5 |
+----------+
1 row in set (0.09 sec)
```

11. Count based on name

```
mysql> select count(name)   from student_231bcada31;
+-------------+
| count(name) |
+-------------+
|           4 |
+-------------+
1 row in set (0.02 sec)
```

Group by statement

1. Create table

```
mysql> create table emp_231bcada31(e_id int, e_name varchar(20),dept varchar(20))
Query OK, 0 rows affected (0.25 sec)
```

2. Insert values

```
mysql> INSERT INTO emp_231bcada31 VALUES('01', 'rohan', 'hr'),('02', 'gagan', 'hr'),('03', 'vineeth', 'marketing'),('04', 'nithin', 'teaching');
Query OK, 4 rows affected (0.07 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

3. Display

```
mysql> select * from emp_231bcada31;
+------+---------+-----------+
| e_id | e_name  | dept      |
+------+---------+-----------+
|    1 | rohan   | hr        |
|    2 | gagan   | hr        |
|    3 | vineeth | marketing |
|    4 | nithin  | teaching  |
+------+---------+-----------+
4 rows in set (0.00 sec)
```

4. Display total no. of emp

```
mysql> select count(*) from emp_231bcada31;
+----------+
| count(*) |
+----------+
|        4 |
+----------+
1 row in set (0.00 sec)
```

5 Display dept and total no. of emp in each dept

```
mysql> select dept,count(e_id) from emp_231bcada31 group by dept;
+-----------+-------------+
| dept      | count(e_id) |
+-----------+-------------+
| hr        |           2 |
| marketing |           1 |
| teaching  |           1 |
+-----------+-------------+
3 rows in set (0.05 sec)
```

6. Display the count of dept grouped by employee id

```
mysql> select count(e_id) from emp_231bcada31 group by dept;
+-------------+
| count(e_id) |
+-------------+
|           2 |
|           1 |
|           1 |
+-------------+
3 rows in set (0.00 sec)
```

5-02-2025

Account master

1.Create table account master

```
mysql> create table acc_master_231bcada31(acc_no int , acc_type varchar(15), bal_amt decimal(10,2), c_name varchar(50), DOC Date, cust_id int);
Query OK, 0 rows affected (0.02 sec)
```

2. Insert 5 records

```
mysql> insert into acc_master_231bcada31
    -> values (121, 'sb', 23456, 'rohan', '2015-01-11', 01),
    ->        (122, 'sd', 12345, 'rekha', '2022-12-12', 02),
    ->        (123, 'rb', 67545, 'gagan', '2016-01-13', 03),
    ->        (124, 'fd', 35833, 'aish', '2023-01-14', 04),
    ->        (125, 'fd', 23662, 'swathi', '2024-01-15', 05);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

3. Show all tables

```
mysql> show tables;
+-----------------------+
| Tables_in_stud        |
+-----------------------+
| acc_master            |
| acc_master_231bcada31 |
| student               |
+-----------------------+
3 rows in set (0.01 sec)
```

4. Describe the structure of table

```
mysql> desc acc_master_231bcada31
    -> ;
+----------+---------------+------+-----+---------+-------+
| Field    | Type          | Null | Key | Default | Extra |
+----------+---------------+------+-----+---------+-------+
| acc_no   | int           | YES  |     | NULL    |       |
| acc_type | varchar(15)   | YES  |     | NULL    |       |
| bal_amt  | decimal(10,2) | YES  |     | NULL    |       |
| c_name   | varchar(50)   | YES  |     | NULL    |       |
| DOC      | date          | YES  |     | NULL    |       |
| cust_id  | int           | YES  |     | NULL    |       |
+----------+---------------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

5. Create a table ac1 that contains acc_no and cust_name with no values

```
mysql> create table acc1 as select acc_no,c_name from acc_master_231bcada31
where 1=0;
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

6. Retrieve all info from account master where acc no is 0112 and acc_type is sb

```
mysql> select * from acc_master_231bcada31 where acc_no= 121 and acc_type = 'sb';
+--------+----------+----------+--------+------------+---------+
| acc_no | acc_type | bal_amt  | c_name | DOC        | cust_id |
+--------+----------+----------+--------+------------+---------+
|    121 | sb       | 23456.00 | rohan  | 2015-01-11 |       1 |
+--------+----------+----------+--------+------------+---------+
1 row in set (0.00 sec)
```

7. Show different types of acc by eliminating the repeated types of acc

```
mysql> select distinct acc_type from acc_master_231bcada31;
+----------+
| acc_type |
+----------+
| sb       |
| sd       |
| rb       |
| fd       |
+----------+
4 rows in set (0.00 sec)
```

8. Display the detail of a customer whose name is 6 char long and starts with char ‘a’

```
mysql> select * from acc_master_231bcada31 where c_name like 'a_____';
Empty set (0.00 sec)
```

9. Arrange data according to descending order of acc_no

```
mysql> select * from acc_master_231bcada31 order by acc_no desc;
+--------+----------+----------+--------+------------+---------+
| acc_no | acc_type | bal_amt  | c_name | DOC        | cust_id |
+--------+----------+----------+--------+------------+---------+
|    125 | fd       | 23662.00 | swathi | 2024-01-15 |       5 |
|    124 | fd       | 35833.00 | aish   | 2023-01-14 |       4 |
|    123 | rb       | 67545.00 | gagan  | 2016-01-13 |       3 |
|    122 | sd       | 12345.00 | rekha  | 2022-12-12 |       2 |
|    121 | sb       | 23456.00 | rohan  | 2015-01-11 |       1 |
+--------+----------+----------+--------+------------+---------+
5 rows in set (0.00 sec)
```

10. Find the total number of accounts created by each user

```
mysql> select c_name,count(acc_type) from acc_master_231bcada31 group by c_name;
+--------+-----------------+
| c_name | count(acc_type) |
+--------+-----------------+
| rohan  |               1 |
| rekha  |               1 |
| gagan  |               1 |
| aish   |               1 |
| swathi |               1 |
+--------+-----------------+
5 rows in set (0.00 sec)
```

11. Find out the customers who is having more than 1 acc

```
mysql> select c_name from acc_master_231bcada31 group by c_name having count(*)>1 ;
Empty set (0.00 sec)
```

12. Find out the number of acc which is created after 3rd jan 2020

```
mysql> select count(*) from acc_master_231bcada31 where DOC>'2020/01/03';
+----------+
| count(*) |
+----------+
|        3 |
+----------+
1 row in set, 1 warning (0.00 sec)
```

13.Demonstrate aggrigation function

```
mysql> select sum(bal_amt), max(bal_amt),min(bal_amt) from acc_master_231bcada31;
+--------------+--------------+--------------+
| sum(bal_amt) | max(bal_amt) | min(bal_amt) |
+--------------+--------------+--------------+
|    162841.00 |     67545.00 |     12345.00 |
+--------------+--------------+--------------+
1 row in set (0.00 sec)
```

20-02-25

Association and members

1.Create database

```
mysql> create table project(pid varchar(10) primary key,pname varchar(20),college varchar(20) defa
ult 'sju',dept varchar(25) not null,research_head varchar(25) not null);
                                                                                  Query OK, 0 ro
ws affected (0.01 sec)
```

2. create table research scholar and project

```
mysql> create table research_scholar(name varchar(25),id varchar(15) primary key)
    -> ;
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> create table research_scholar(name varchar(25),id varchar(15) primary key,pid varchar(20),
foreign key (pid) references project(pid));
Query OK, 0 rows affected (0.04 sec)
```

3. insert values into both the tables

```
mysql> insert into project values (1,'DBMS','sjc','BIGData','jeshma'),(2,'Python','sju','IT','Aron
');
Query OK, 2 rows affected (0.00 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

```
mysql> insert into research_scholar values('Rohan','bcada31',1),('Gagan','bcada32',2),('Rekha','bc
ada34',1);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

4. join the tables

```
mysql> select * from project p, research_scholar r where p.pid=r.pid;
+-----+--------+---------+---------+---------------+-------+---------+------+
| pid | pname  | college | dept    | research_head | name  | id      | pid  |
+-----+--------+---------+---------+---------------+-------+---------+------+
| 1   | DBMS   | sjc     | BIGData | jeshma        | Rohan | bcada31 | 1    |
| 1   | DBMS   | sjc     | BIGData | jeshma        | Rekha | bcada34 | 1    |
| 2   | Python | sju     | IT      | Aron          | Gagan | bcada32 | 2    |
+-----+--------+---------+---------+---------------+-------+---------+------+
3 rows in set (0.00 sec)
```

5. create table association and members

create table association (aid int primary key, aname varchar(15));

```
mysql>  create table ass_231bcada31(a_id int primary key,a_name varchar(20));
Query OK, 0 rows affected (0.02 sec)

mysql> create table member(m_id int primary key,a_id int, foreign key (a_id) references ass_231bca
da31(a_id));
Query OK, 0 rows affected (0.04 sec)
```

6. insert values into both tables

```
mysql> insert into member values(12,1,'Rohan'),(13,2,'rohith'),(15,1,'vineeth');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

7. display both tables

```
mysql> select * from ass_231bcada31;
+------+-----------+
| a_id | a_name    |
+------+-----------+
|    1 | Analytica |
|    2 | DataGram  |
+------+-----------+
2 rows in set (0.00 sec)

mysql> select * from member;
+------+------+---------+
| m_id | a_id | m_name  |
+------+------+---------+
|   12 |    1 | Rohan   |
|   13 |    2 | rohith  |
|   15 |    1 | vineeth |
+------+------+---------+
3 rows in set (0.00 sec)
```

8. describe both tables

```
mysql> desc ass_231bcada31 ;
+--------+-------------+------+-----+---------+-------+
| Field  | Type        | Null | Key | Default | Extra |
+--------+-------------+------+-----+---------+-------+
| a_id   | int         | NO   | PRI | NULL    |       |
| a_name | varchar(20) | YES  |     | NULL    |       |
+--------+-------------+------+-----+---------+-------+
2 rows in set (0.00 sec)

mysql> desc member ;
+--------+-------------+------+-----+---------+-------+
| Field  | Type        | Null | Key | Default | Extra |
+--------+-------------+------+-----+---------+-------+
| m_id   | int         | NO   | PRI | NULL    |       |
| a_id   | int         | YES  | MUL | NULL    |       |
| m_name | varchar(20) | YES  |     | NULL    |       |
+--------+-------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

9. join the tables

```
mysql> select * from ass_231bcada31, member where ass_231bcada31.a_id=member.a_id;
+------+-----------+------+------+---------+
| a_id | a_name    | m_id | a_id | m_name  |
+------+-----------+------+------+---------+
|    1 | Analytica |   12 |    1 | Rohan   |
|    1 | Analytica |   15 |    1 | vineeth |
|    2 | DataGram  |   13 |    2 | rohith  |
+------+-----------+------+------+---------+
3 rows in set (0.00 sec)
```

10. whether column name of primary key should be same as column name in foreign key

No, it need not be the same as we can change the column name

```
mysql> create table example(id int primary key, associaton_id int, foreign key (associaton_id) references ass_231bcada31(a_id));
Query OK, 0 rows affected (0.04 sec)

mysql>
```

11. create short form and access the table

```
mysql> select a.a_name, m.m_id,m.m_name  from ass_231bcada31 a, member m where a.a_id=m.a_id;
+-----------+------+---------+
| a_name    | m_id | m_name  |
+-----------+------+---------+
| Analytica |   12 | Rohan   |
| Analytica |   15 | vineeth |
| DataGram  |   13 | rohith  |
+-----------+------+---------+
3 rows in set (0.00 sec)
```

12.create a table and demonstrate Default constraint

```
mysql> create table student_31(id int primary key, name varchar(20),class varchar(20) default '10th');
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> insert into student_31(id,name) values (31,'rohan'),(32,'gagan'),(22,'roh') ;
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from student_31;
+----+-------+-------+
| id | name  | class |
+----+-------+-------+
| 22 | roh   | 10th  |
| 31 | rohan | 10th  |
| 32 | gagan | 10th  |
+----+-------+-------+
3 rows in set (0.00 sec)
```

13 . overwrite a default value and display

```
mysql> insert into student_31 values (33,'rekha','12th') ;
Query OK, 1 row affected (0.01 sec)

mysql> select * from student_31;
+----+-------+-------+
| id | name  | class |
+----+-------+-------+
| 22 | roh   | 10th  |
| 31 | rohan | 10th  |
| 32 | gagan | 10th  |
| 33 | rekha | 12th  |
+----+-------+-------+
4 rows in set (0.00 sec)
```

```
mysql> alter table student_31 alter class drop default;
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Check constraints

1.Create table employee such that employees should only belong to Bengaluru or mysore or manglore.

```
mysql> create table employee_231bcada31( emp_id int primary key, emp_name varchar(20) not null ,location varchar(20), check (location in ("bengaluru", "myso
re" "mangaluru")));
Query OK, 0 rows affected (0.01 sec)
```

2.Create table employee such that employees name must start with a

```
mysql> create table emp(emp_id int primary key, emp_name varchar(20) not null, check (emp_name like "a_____"));
Query OK, 0 rows affected (0.03 sec)
```

3.Add check contrain to existing table

```
mysql> create table student_31(id int primary key, name varchar(20),class varchar(20) default '10th');
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> alter table student_31 add check (age>18);
Query OK, 4 rows affected (0.07 sec)
Records: 4  Duplicates: 0  Warnings: 0
```